

Mohammed-Khalil Ghali, PhD Candidate

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EDUCATION

Binghamton University, School of Systems Science and Industrial Engineering, SUNY **Binghamton, USA**
Doctor of Philosophy in Industrial and Systems Engineering **Expected December 2026**
Dissertation: Unified Intelligence: Orchestrating Machine Learning, Deep Learning and Agentic Generative AI in Complex Multidomain Systems

Binghamton University, School of Systems Science and Industrial Engineering, SUNY **Binghamton, USA**
Master of Science in Industrial and Systems Engineering (GPA: 3.83) **May 2024**
Thesis: Leveraging Generative AI and in Context Learning to Reshape Human-Text Interaction: A Novel Paradigm for Information Retrieval, Named Entities Extraction, and Database Querying

Al Akhawayn University **Ifrane, Morocco**
Bachelor of Science in Computer Science (GPA: 3.55) **June 2022**

RESEARCH EXPERIENCE

Meta Platforms, Inc. **Menlo Park, USA**
AI Engineer - Contractor **February 2026 – Present**

- Designed and deployed a full stack multi-agent AI engine for automated hardware test plan generation orchestrating 4+ specialized AI agents end-to-end reducing test plan creation and validation from days of engineering effort to minutes
- Productionized an integrated multi-agent AI test cases verdict recommender as scheduled FaaS serverless jobs with a durable work queue, data persistence, and continuous self-discovering resumable batch scoring, cutting hardware repair-workflow tests reviewing from hours of manual log analysis to minutes of auditable, in-workflow recommendations
- Created a suite of 15+ productivity AI agents skills bundled into a unified plugin to automate repetitive workflows eliminating hours of operational work and driving adoption of agentic AI as a force multiplier for our org's engineering productivity
- Built a cross-platform agents converter enabling seamless migration of AI agents capabilities across multiple generative AI platforms and harnesses, ensuring tool portability and reuse allowing teams to adopt new AI platforms without rebuilding their agentic workflows from scratch
- Architecting the technical roadmap of a unified AI-native platform for our org incorporating skills, plugins, agents, tools, MCP servers, and reusable no-ops deployment templates reducing time-to-production while enabling SMEs to ship production-ready AI without infrastructure expertise

Lattice Semiconductor **Hillsboro, USA**
AI Engineer - Contractor **December 2023 – February 2026**

- Participated in the development and maintenance of a cloud hosted data analytics platform handling data from more than 13 internal and external sources transformed to 100+ tables with 30B rows of ops data
- Delivered 4 fully cloud-native quality apps to drive and encourage data-informed decision making
- Developed a modular MLOps product (The MLPlatform) that standardizes data preparation, model training, deployment, inference, and monitoring to accelerate ML projects delivery
- Developed and deployed end-to-end deep learning models using the MLPlatform for defect analysis presented in a web application for foundry engineers
- Contributed to 2.99% average yield improvement across FPGA product families (ranging 0.36%-10.61%) and reduced customer returns by 141,000+ potentially failing dies through the data analytics platform
- Supported the optimization of testing equipment utilization to support the company's new high volume production initiative
- Developed an agentic generative AI solution for QA customer requests, cutting ticket resolution time from 1+ month to minutes

Watson Institute of Systems Excellence **Binghamton, NY**
Graduate Research Associate: LockerRoom345 **August 2022 – December 2023**

- Contributed to the end-to-end development and launch of LockerRoom345 website, a charity platform facilitating the provisioning of essential clothing to 2000+ Binghamton area students in need with 19,361 items fulfilled yearly through Dick's Sporting Goods donations (www.lockerroom345.org)

SCHOLARSHIPS & AWARDS

Merit Based Scholarships

- Received \$200,000+ in research assistantships, Watson Institute of Systems Excellence, Binghamton University
- Awarded \$8,000+ in doctoral student grants, Binghamton University
- Earned a \$20,000 undergraduate excellence and merit scholarship, Al Akhawayn University

Awards

- Awarded \$10,000+ in competitive funding through grants for conference attendance and academic recognitions
- 1st place: Entrepreneurial challenge at the international summer school hosted by the American University of Sharjah (UAE) competing against students from NUS, University of Manchester and Columbia University
- 1st place: Best Master's thesis competition at IEOM World Congress 2024, Detroit, Michigan, USA
- 2nd place: DATA4GOOD generative AI/data science competition by Purdue University, Informa and Microsoft
- Finalist: Doctoral colloquium pitch competition at the IISE 2025 Annual Conference
- Finalist: IISE 2024 Data Analytics and Information Systems (DAIS) analytics web app competition

PUBLICATIONS

- **Ghali M.K.**, Farrag A, Won D, Jin Y, "Enhancing Knowledge Retrieval with In-Context Learning and Semantic Search through Generative AI", *Knowledge-Based Systems*, 2025: 113047. (Published)
- Akino K, **Ghali M.K.**, Ijuin H, Yamada T, Yoon S.W, "Demand and Capacity Sharing Decisions and Protocols in Multi-microgrid Using Deep Learning", *FAIM International Conference*, 2025 (Published)
- Farrag A, Cao N, **Ghali M.K.**, Jin Y, Won D, "BURGAL: An Iterative Guided Active Learning Sampling Approach for Smart Manufacturing Systems", *International Journal of Advanced Manufacturing Technologies*, 2025 (Published)
- El Baz H, **Ghali M.K.**, Andrus L, Won D, Yoon S.W, Wang Y, "Generative AI-based Optimization for High-Mix Warehouses: Integrating Agentic LLMs and DES", *Computers & Industrial Engineering*, 2025 (Published)
- El Baz H, **Ghali M.K.**, Khalid U, Pandey V, Won D, Yoon S.W, Wang Y, "A Tool-Grounded Multi-Agent LLM Workflow for Manufacturing Line Decision Support", *Computers & Industrial Engineering*, 2026 (Under review)
- **Ghali M.K.**, Farrag A, Sakai H, El Baz H, Lam S, Jin Y, "GAMedX: Generative AI-based Medical Entity Data Extractor using Large Language Models", *Journal of Healthcare Informatics Research*, 2025 (Under review)
- **Ghali M.K.**, Farrag A, Lam S, Won D, "BEYONDWORDS is All You Need: Agentic Generative AI based Social Media Themes Extractor", *Online Social Networks and Media Journal*, 2025 (Under review)
- **Ghali M.K.**, Molina O, Gershenson Garcia C, Won D, "Forecasting Commodity Price Shocks Using Temporal and Semantic integration of Prices Signals and Agentic Generative AI Extracted Economic News", *IEEE Transactions on Knowledge and Data Engineering*, 2026 (Under review)
- **Ghali M.K.**, Tamilarasan K, La T, Won D, "Predicting Static Leakage Yield Loss During Wafer Sort for Semiconductors Supply Planning using Wafer Acceptance Test Data", *International Journal of Production Research*, 2026 (Under review)
- **Ghali M.K.**, Kulkarni S, Kulkarni R, Kulkarni R, Yoon S.W, Won D, "Automated Hardware Validation Test Plan Generation for Large Scale AI Datacenter Platforms Using a Generative AI Multi-Agents Architecture", *IEEE Transactions on Reliability*, 2026 (Under review)
- Jaradat H, **Ghali M.K.**, Kulkarni S, Kulkarni R, Yoon S.W "An LLM-Orchestrated Neuro-Symbolic Framework for Verifying Diagnostic Signal Reachability in Electronic Systems", *IEEE Transactions on Reliability*, 2026 (Under Internal Review)

TECHNICAL & RELEVANT SKILLS

Computer Tools and Technologies:

- **Cloud & Infrastructure/DevOps & Monitoring:** AWS, Azure, GCP, Kubernetes, Tupperware, ArgoCD/ArgoWorkflows, Sandcastle, Conveyor, Docker, Terraform, GitHub, Jira, Sentry, Sapling
- **Data & Analytics:** PostgreSQL, Snowflake, Databricks, Tableau, Streamlit
- **ML/DL, Generative AI & MLOps:** LangChain, HuggingFace, TensorFlow, PyTorch, MLflow, OpenAI Agents SDK, Google Agents SDK, Anthropic Agents SDK, LangSmith, MCP, A2A
- **Programming:** Python

Certifications:

- Build, Train, and Deploy ML Pipelines in AWS: Coursera
- Generative AI with Large Language Models in AWS: Coursera
- AZ-900: Microsoft Azure Fundamentals: Microsoft

Languages: Arabic: Native, English: Fluent, French: Fluent